

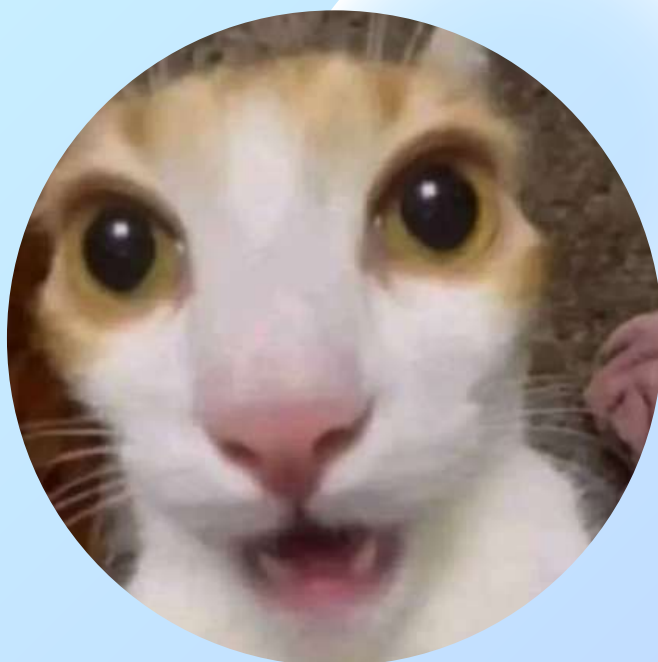
FINAL PROJECT

CENTRAL VIETNAMESE ACCENTED SPEECH RECOGNITION

SEAS 2025

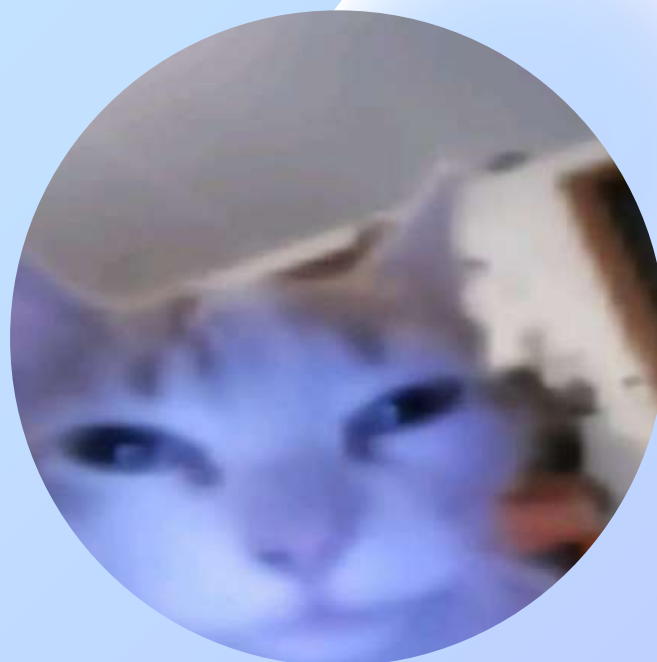
Quang Tri, Friday, August 1st, 2025

MEMBERS



QUANG HUY

Programmer



ĐÌNH QUÂN

Researcher



NGUYỆT TÂM

Researcher

MEMBERS



THANH ĐẠT

Programmer



VIỆT HƯNG

Programmer



MINH ANH

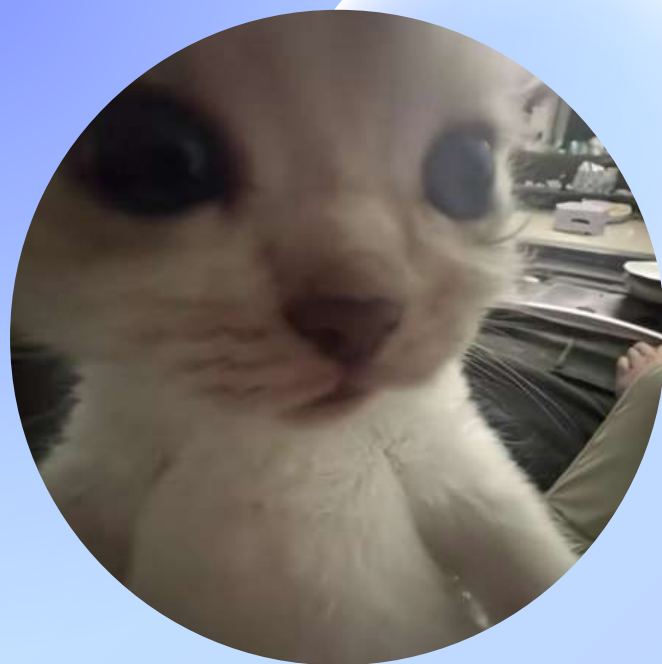
Researcher

ADVISORS



THÀNH LUÂN

Leader



KHÔI NGUYỄN

Advisor



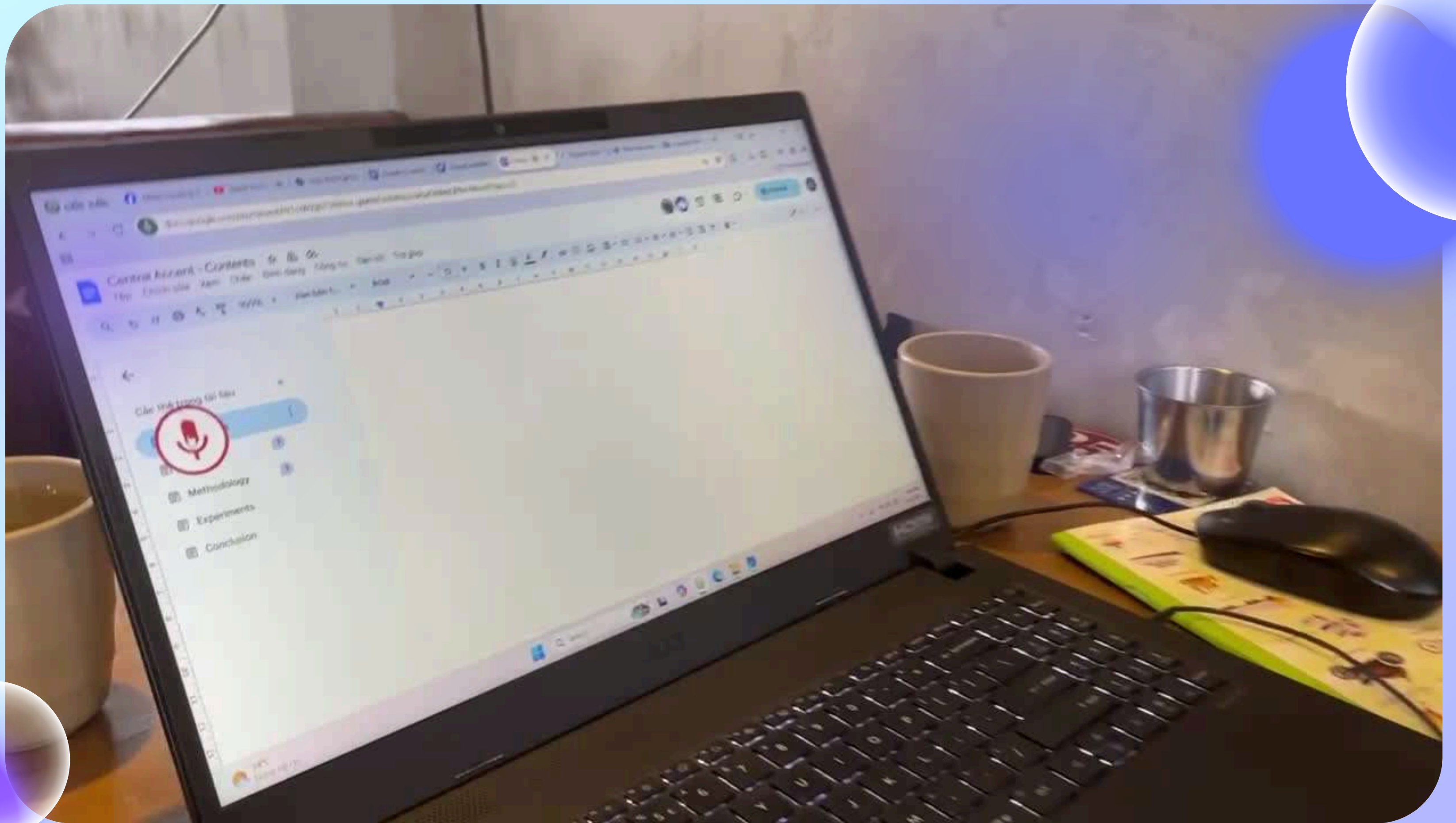
CẢNH HOÀNG

Advisor



BÌNH MINH

Advisor



ChaoCanhWHISPER



**RELATED
WORKS**

**DATA
ANALYSIS**



TABLE OF CONTENTS



METHODOLOGY

EXPERIMENTS

RELATED WORKS

SPEECH PROCESSING

DEFINITION

Speech processing:
analyzing,
modifying, and
synthesizing
speech signals
using **DSP**

MAIN APPLICATIONS

RECOGNITION

CODING

SYNTHESIZE

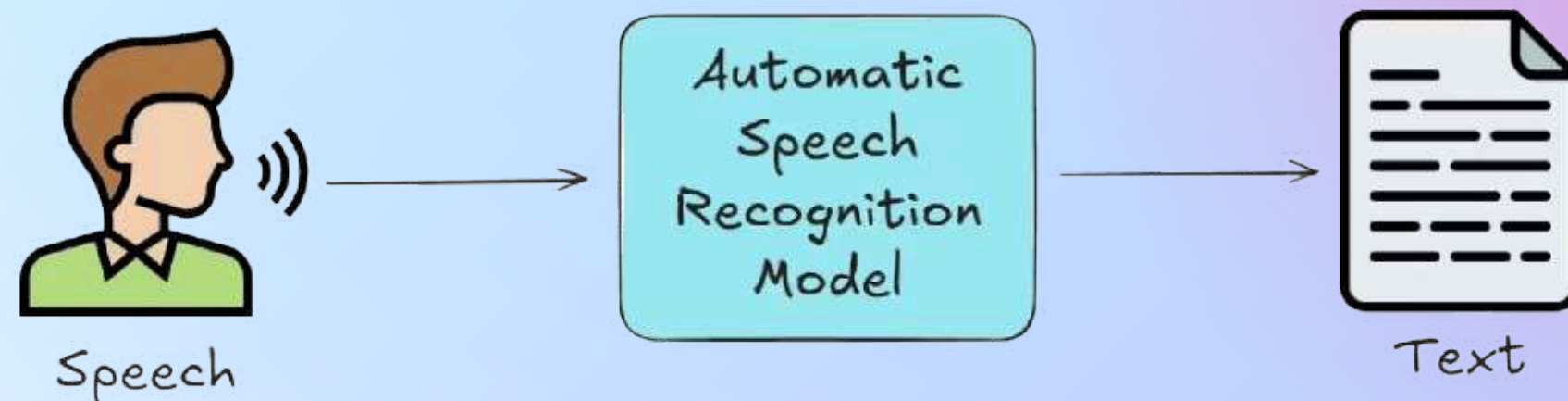
RELATED WORKS

AUTOMATIC SPEECH RECOGNITION (ASR)

DEFINITION

ASR: a technology for automatic voice recognition, a.k.a. speech-to-text conversion

Common approach: Self-Supervised Learning



Popular models: Wav2Vec2, HuBERT,...

EXISTING LIMITATIONS IN VIETNAMESE ASR

- ✗ Data is **not diverse** enough
- ✗ Current models cannot encompass **dialects' complexity**

DATA ANALYSIS

Dataset: **ViMD**

What is the
dataset about?

DATA PREPROCESSING FOR TRAINING:



Resampled: 48kHz → 16kHz



Compressed to float16 format

DATASET OVERVIEW

Key stats:

- 102.56 hours of audio
- ~19,000 utterances
- 1.2 million+ words in transcripts

Data from 63 provinces
of Vietnam (before the
merger of
administrative units)

- Average duration: ~19.11 seconds.
- Total duration: ~24.98 hours.
- Metadata info:

- Entries: 4,705 (0 to 4,704)
- Data columns:

	region	province_code	province_name	filename	text	speakerID	gender	audio
0	Central	36	ThanhHoa	36_0001.wav	Rất là tiện đấy ạ. thí dụ như là tôi muốn về t...	spk_36_0001	0	{array: [-0.004215, -0.007328, -0.005314, -0...
1	Central	36	ThanhHoa	36_0002.wav	Kiến nghị với các cơ quan chức năng nhà nước c...	spk_36_0002	1	{array: [0.001575, 0.00314, 0.00294, 0.00301...
2	Central	36	ThanhHoa	36_0003.wav	Mình cũng đề nghị với các cấp các ngành tìm ra...	spk_36_0003	1	{array: [-0.001329, -0.002783, 0.000412, 0.0...
3	Central	36	ThanhHoa	36_0004.wav	Hiện nay, thì một số cơ sở dịch vụ thẩm mỹ hoặ...	spk_36_0004	1	{array: [0.012955, 0.00402, -0.01261, -0.010...
4	Central	36	ThanhHoa	36_0005.wav	Tuy nhiên đầu đó cũng đang còn chưa dứt điểm B...	spk_36_0004	1	{array: [0.001908, 0.003462, 0.002922, 0.002...

- Dtypes: int64(2), object(6)
- Memory usage: 294.2+ KB
- **Text Analyze:**
 - **Number of unique words: 7,909 words**
 - **Average length: 64 words**

BASIC DATA ANALYSIS

on “TRAIN” split – “CENTRAL” chunk

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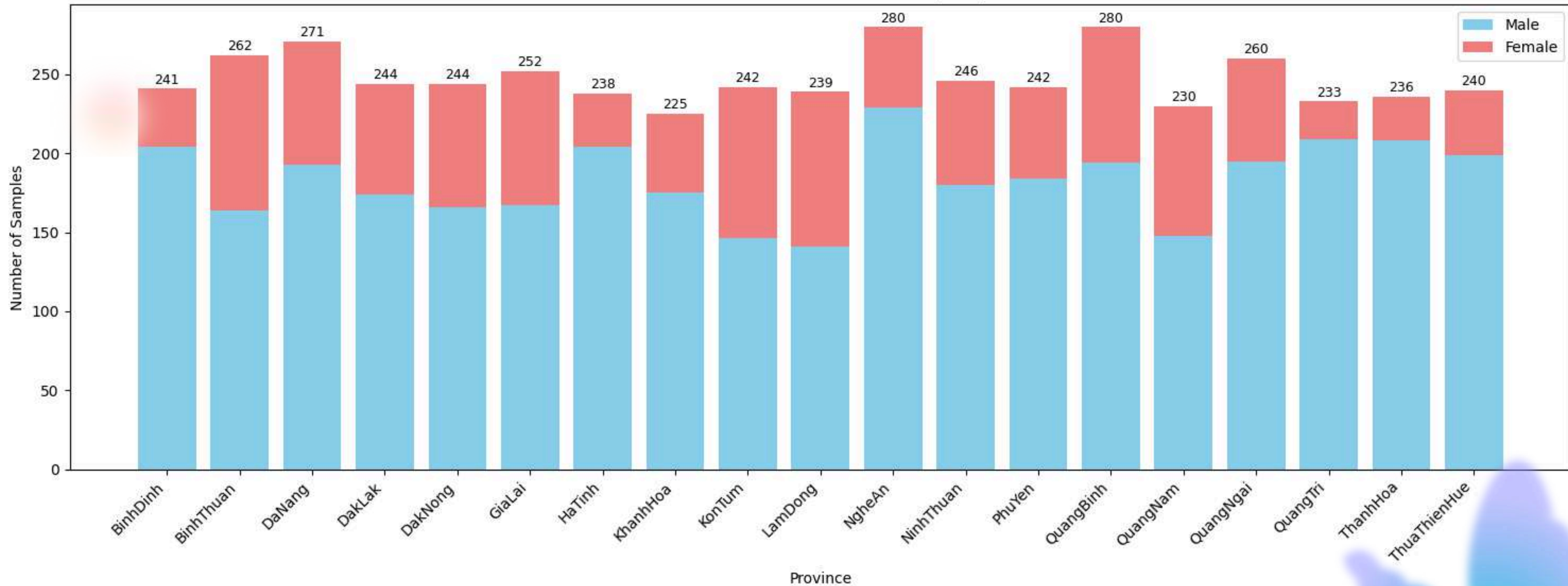
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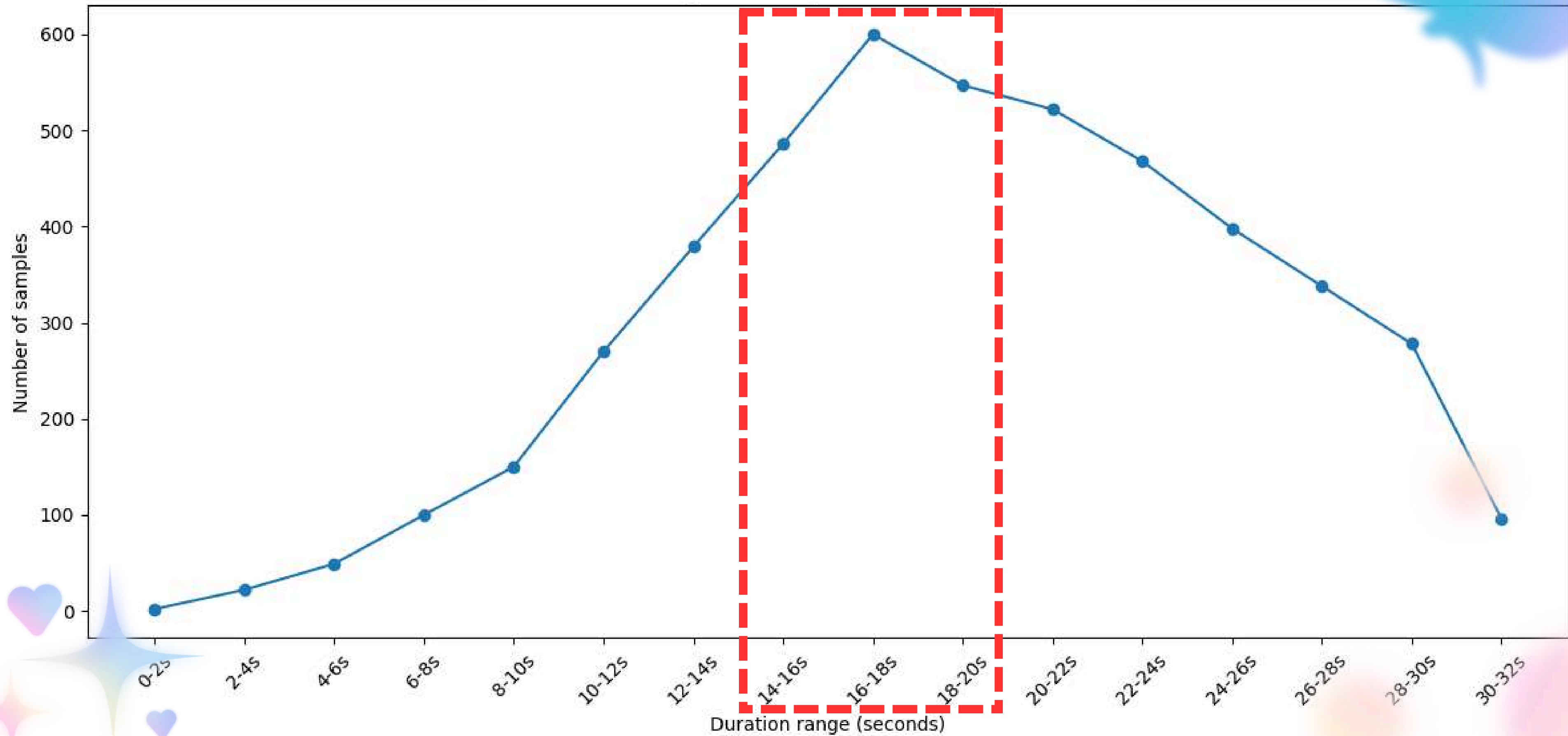
BASIC DATA ANALYSIS

on “TRAIN” split – “CENTRAL” chunk

Distribution of Male and Female Samples per Province

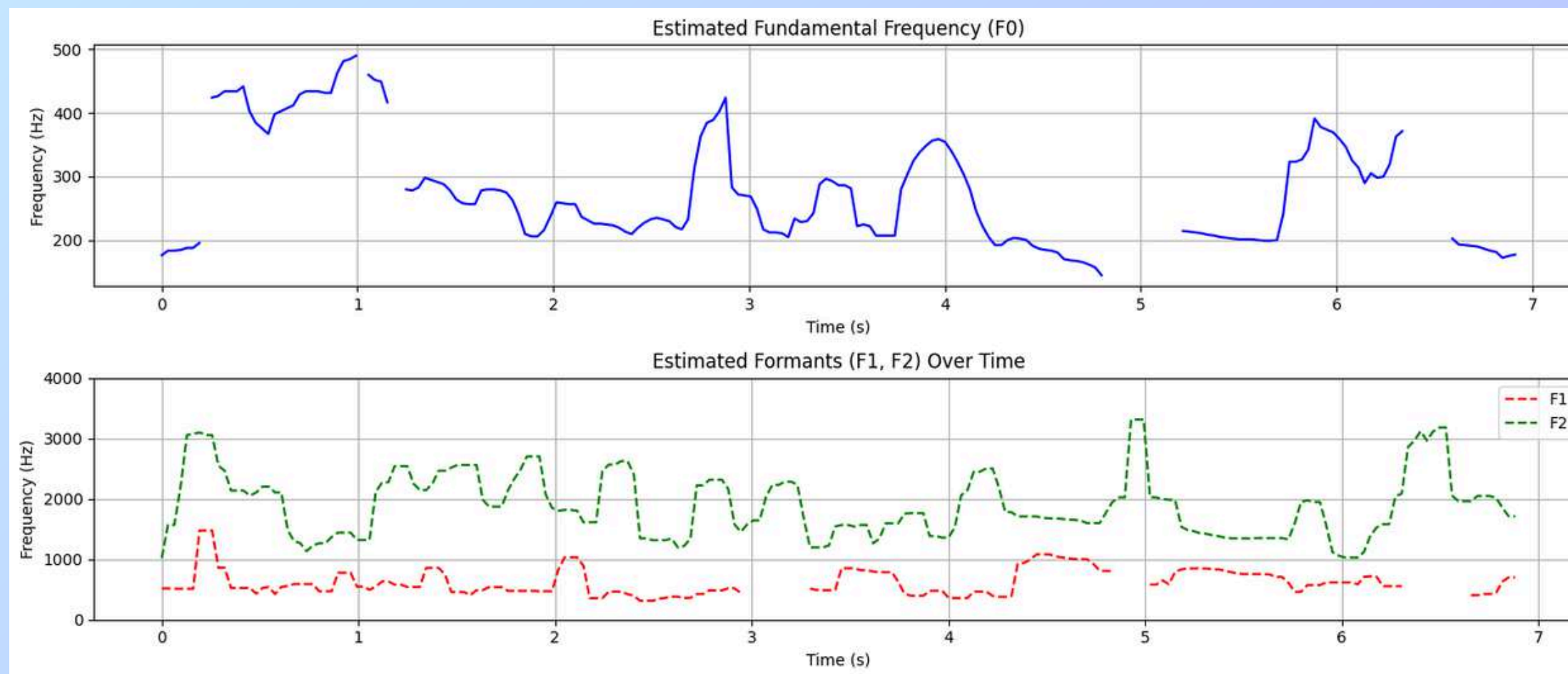
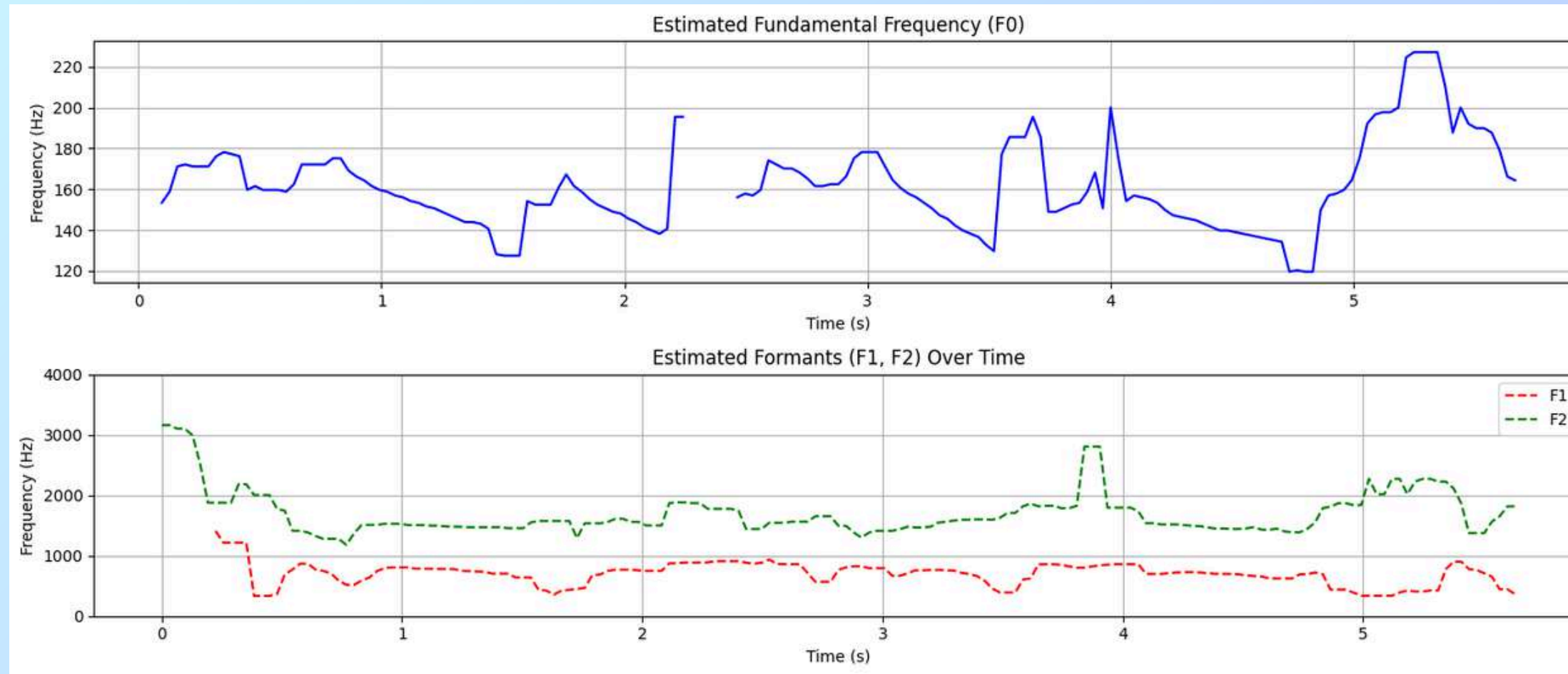


Sample counts per duration range



F0, F1, F2 Analysis

Male

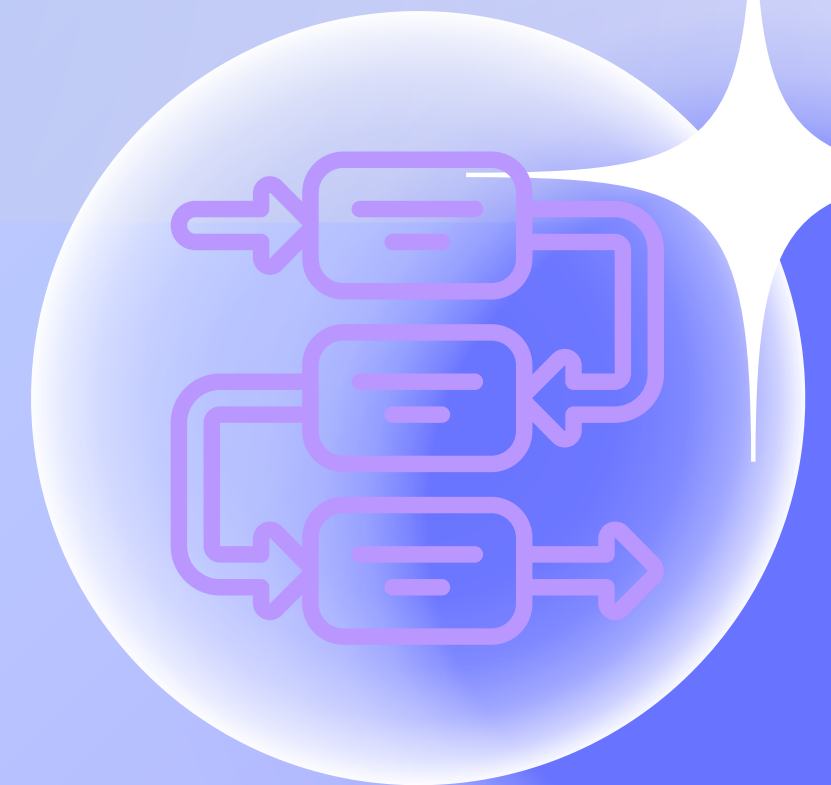


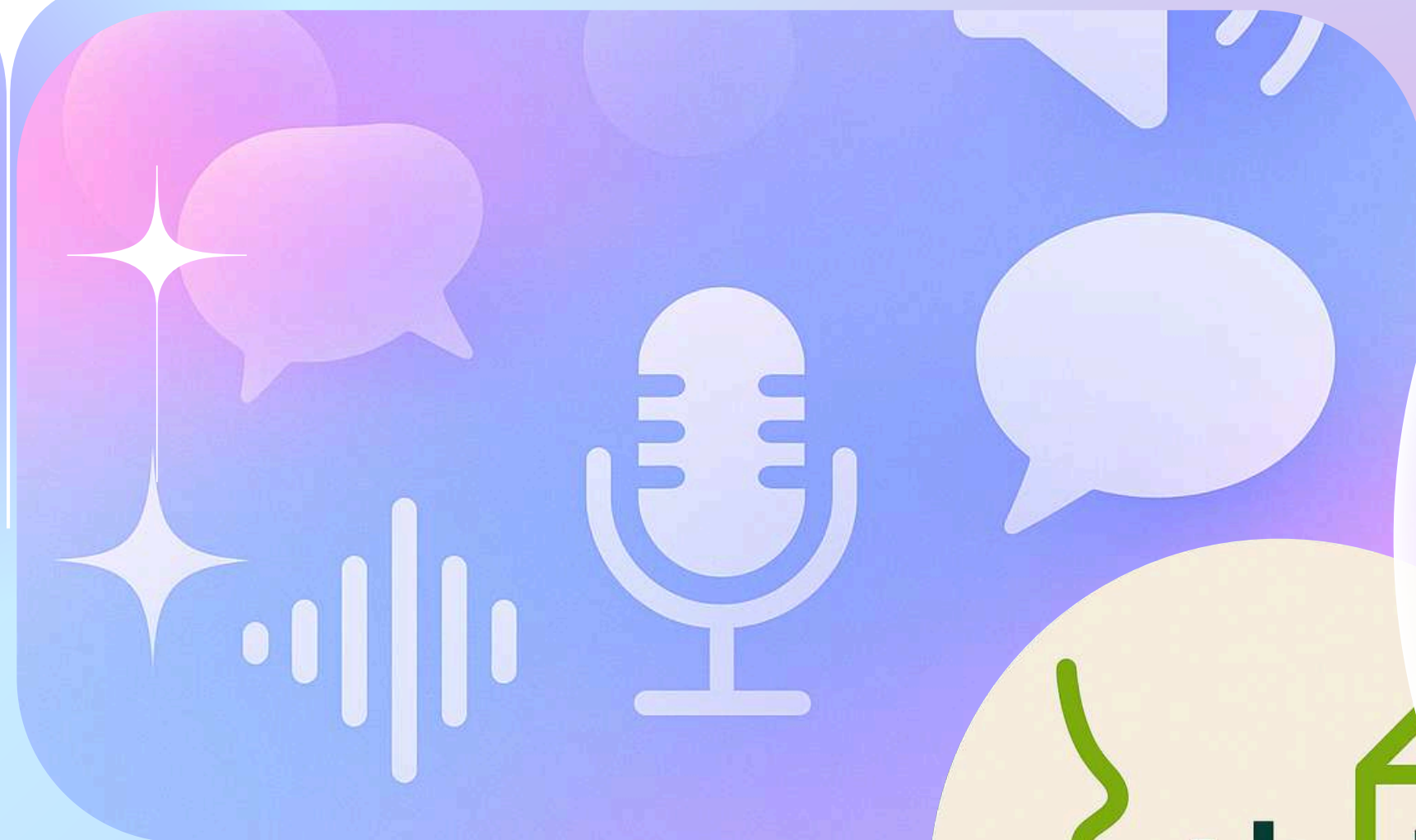
Female



METHODOLOGY

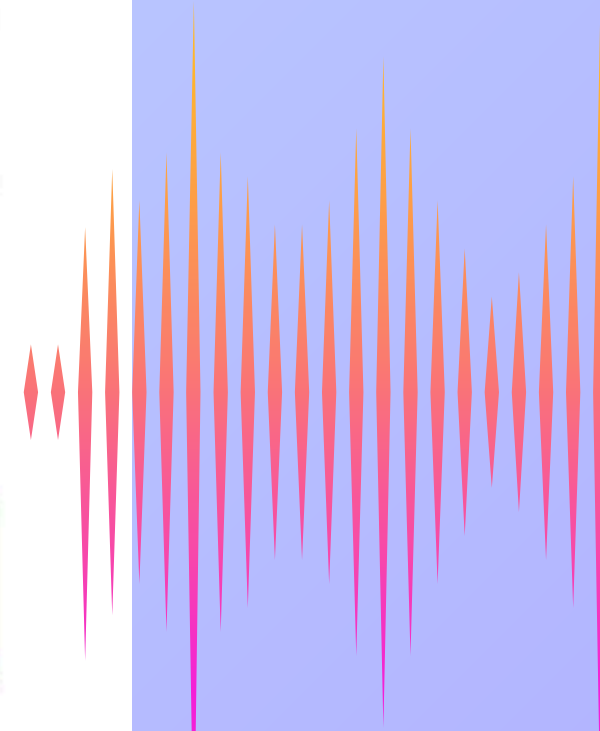
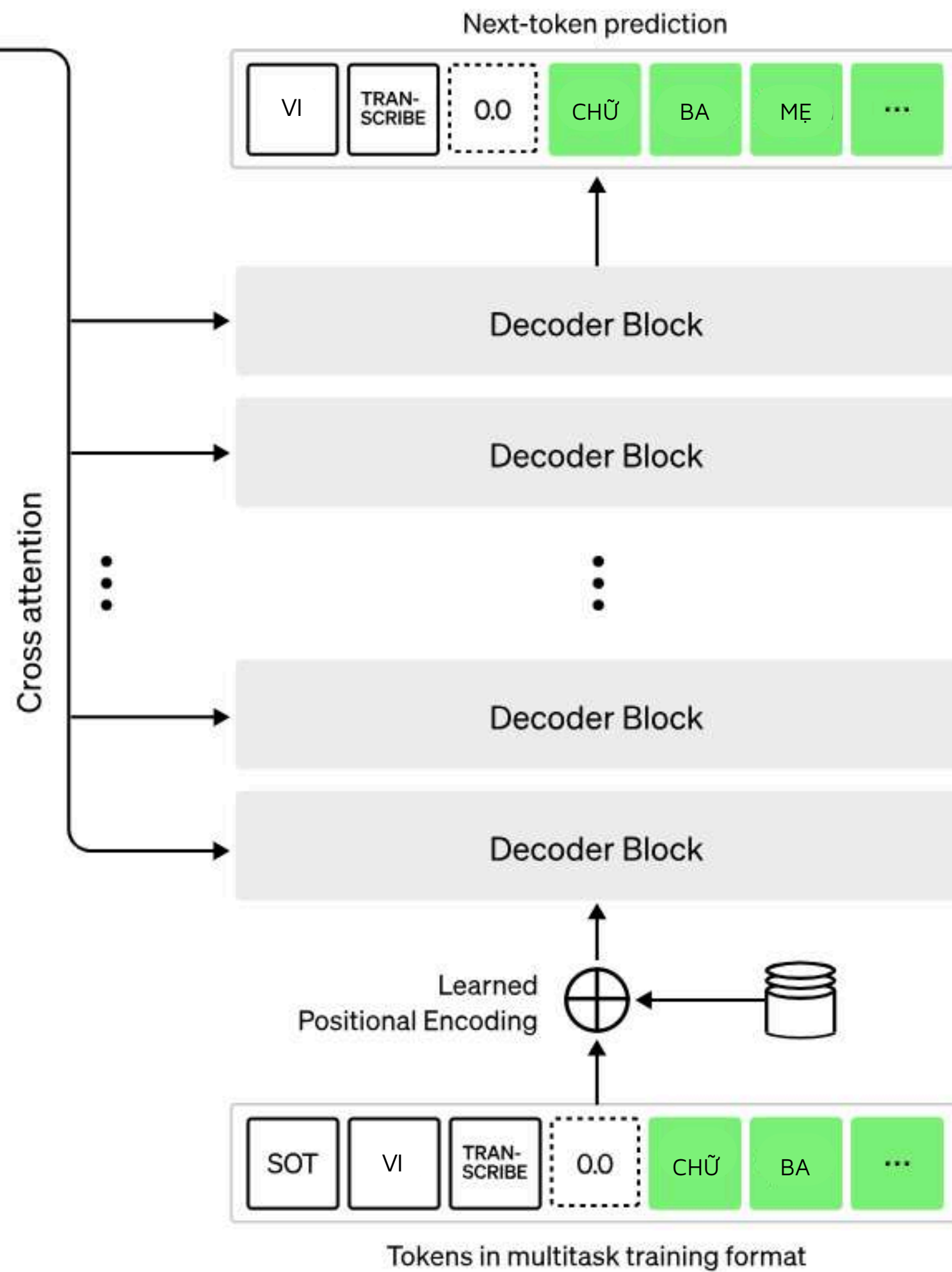
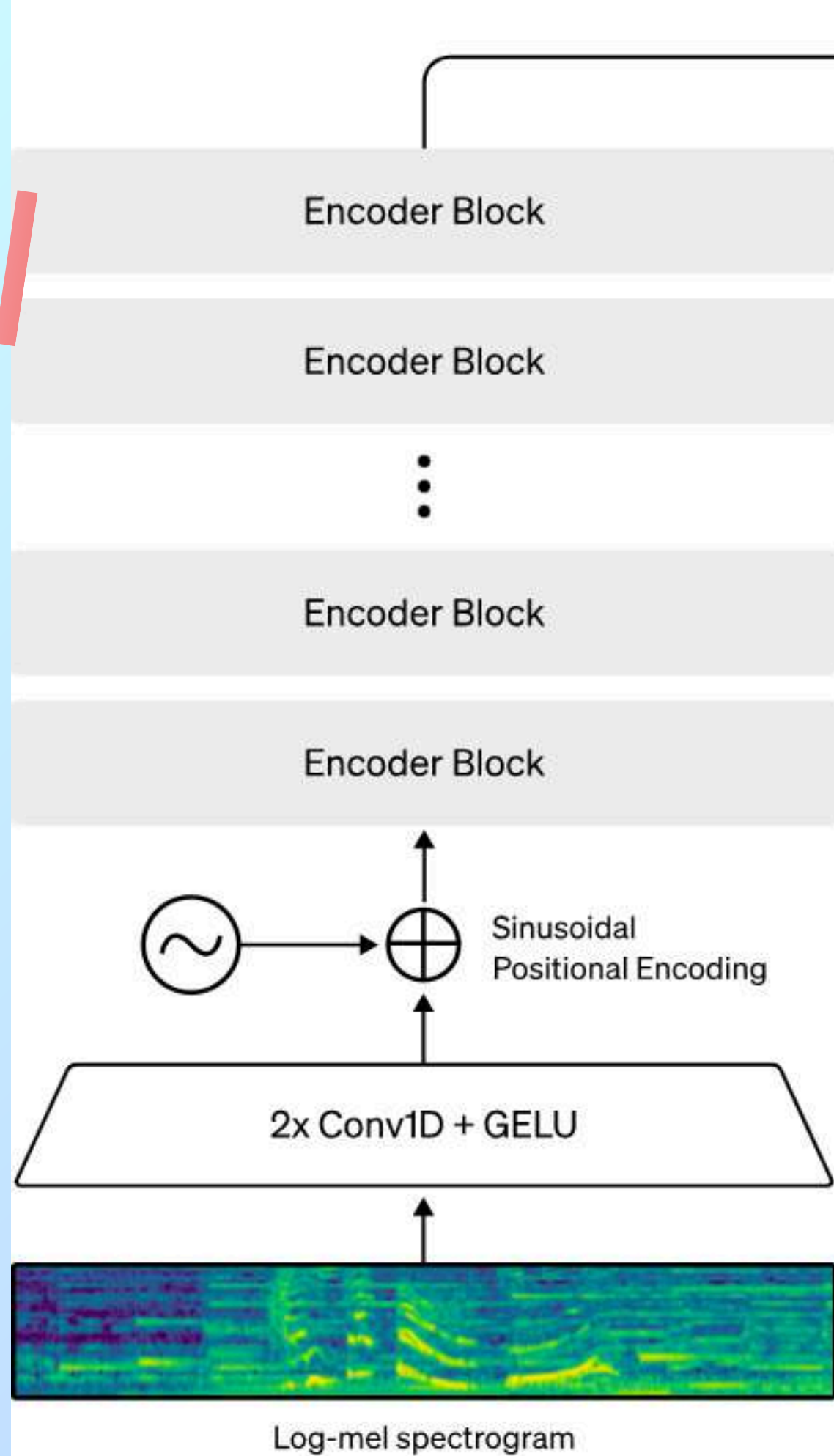
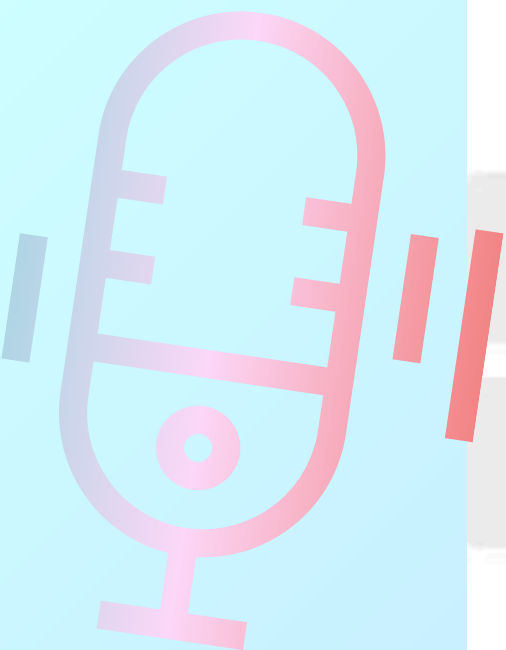
Fine-tuned the PhoWhisper model on a subset of the ViMD dataset





PhoWhisper

Optimized ASR for
Vietnamese





MODEL VARIANTS

PhoWhisper offers five scaled versions:
PhoWhispertiny, **PhoWhisper-base**,
PhoWhispersmall , PhoWhispermedium and
PhoWhisperlarge





EXPERIMENTS

- 
1. Settings
 2. Model Fine-tuning
 - 2.1. Data preparation
 - 2.2. Fine-tuning process
 3. Experimental results
 4. Result analysis & discussion
- 

1. SETTINGS

MODEL

PhoWhisper-base
74M

DATASET

Central dialect in ViMD
dataset, in three parts:
train, valid and test

ENVIRONMENT

Google Colab GPU T4

EVALUATION WER

Based on wrong, missing
and redundant words



WORD ERROR RATE

CATEGORY	WER	NOTES
Human performance	< 5-6%	What human transcribers typically achieve around on clean speech
State-of-the-art ASR	5-20%	<10% on clean audio (e.g. Whisper) closer to 15-20% on noisy or accented input
Challenging tasks	20-40%	Includes conversational speech, noisy environment, low-resource languages
Poor ASR performance	> 40%	Seen in mismatched domains, very noisy input, or undertrained models

2. MODEL FINE-TUNING

DATA PREPARATION

- 3 chunks: **North**, **CENTRAL** and **South**.
- 3 splits: **train**, **valid**, **test**
- **CENTRAL** chunk:
 - + Train: 4,705 audios
 - + Valid: 602 audios
 - + Test: 623 audios

FINE-TUNING PROCESS

- **Load** PhoWhisper pre-trained model
- **Extract** features
- **Tokenize** textual transcripts
- Group samples into **batches** to feed the model
- Set up **parameters** and **start** the training process

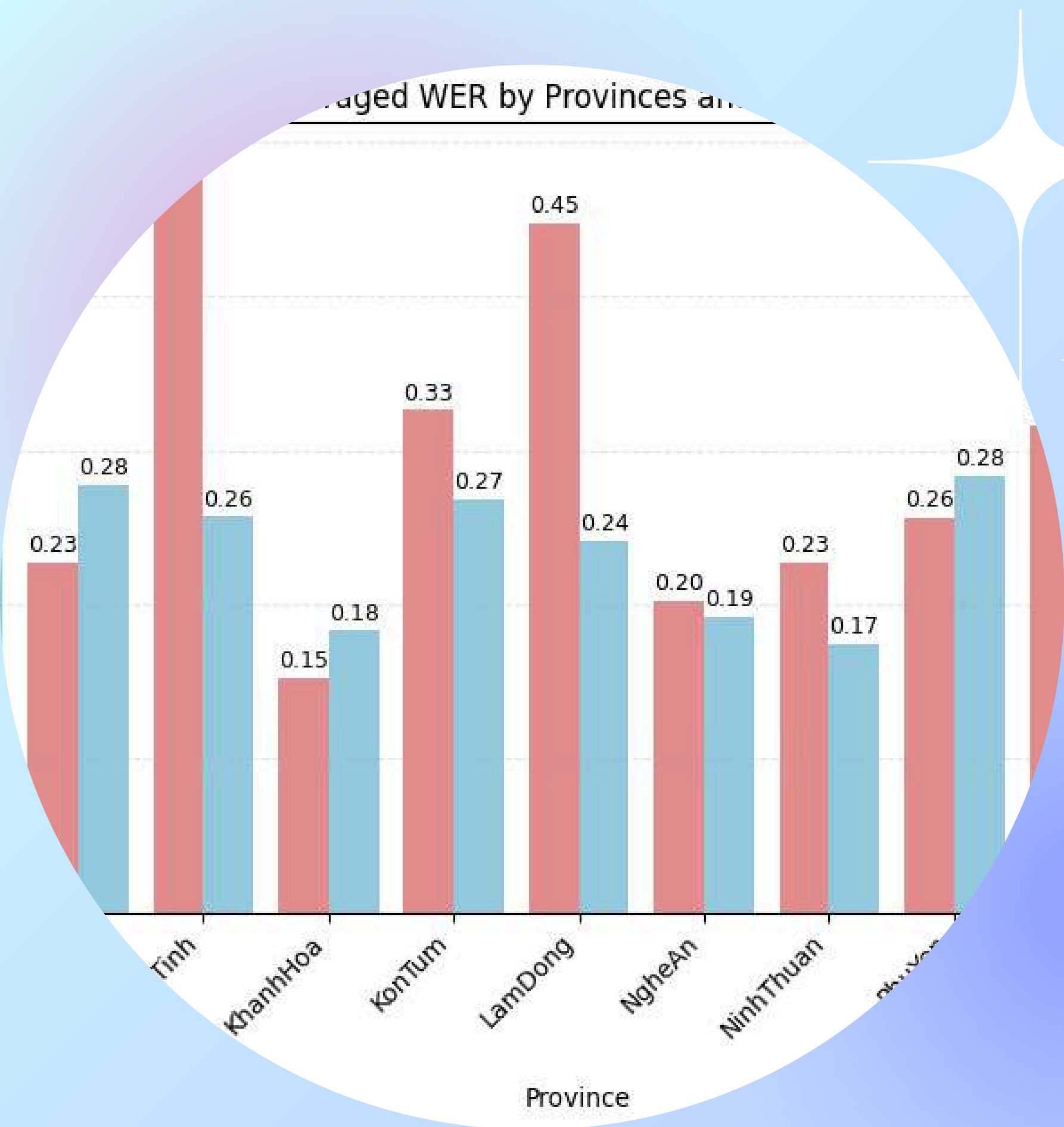


3. EXPERIMENTAL RESULTS

WER ON TEST: **0.2499** | VALIDATION: **0.2587**

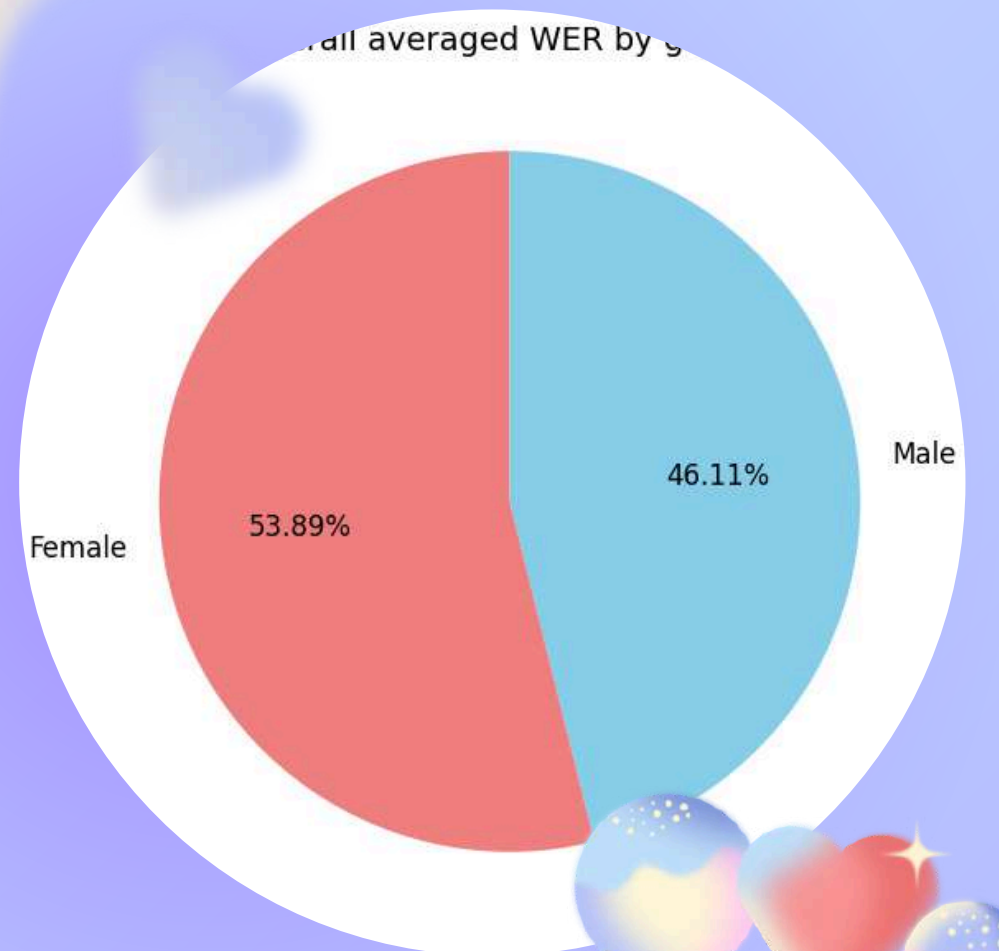
✓ *MEET THE EXPECTATIONS*

ChaoCanhWhisper is a fine-tuned ASR model specifically designed to better recognize Vietnamese Central dialects. Through careful data curation and targeted training, it achieves low word error rates while preserving the nuances of Central dialects.

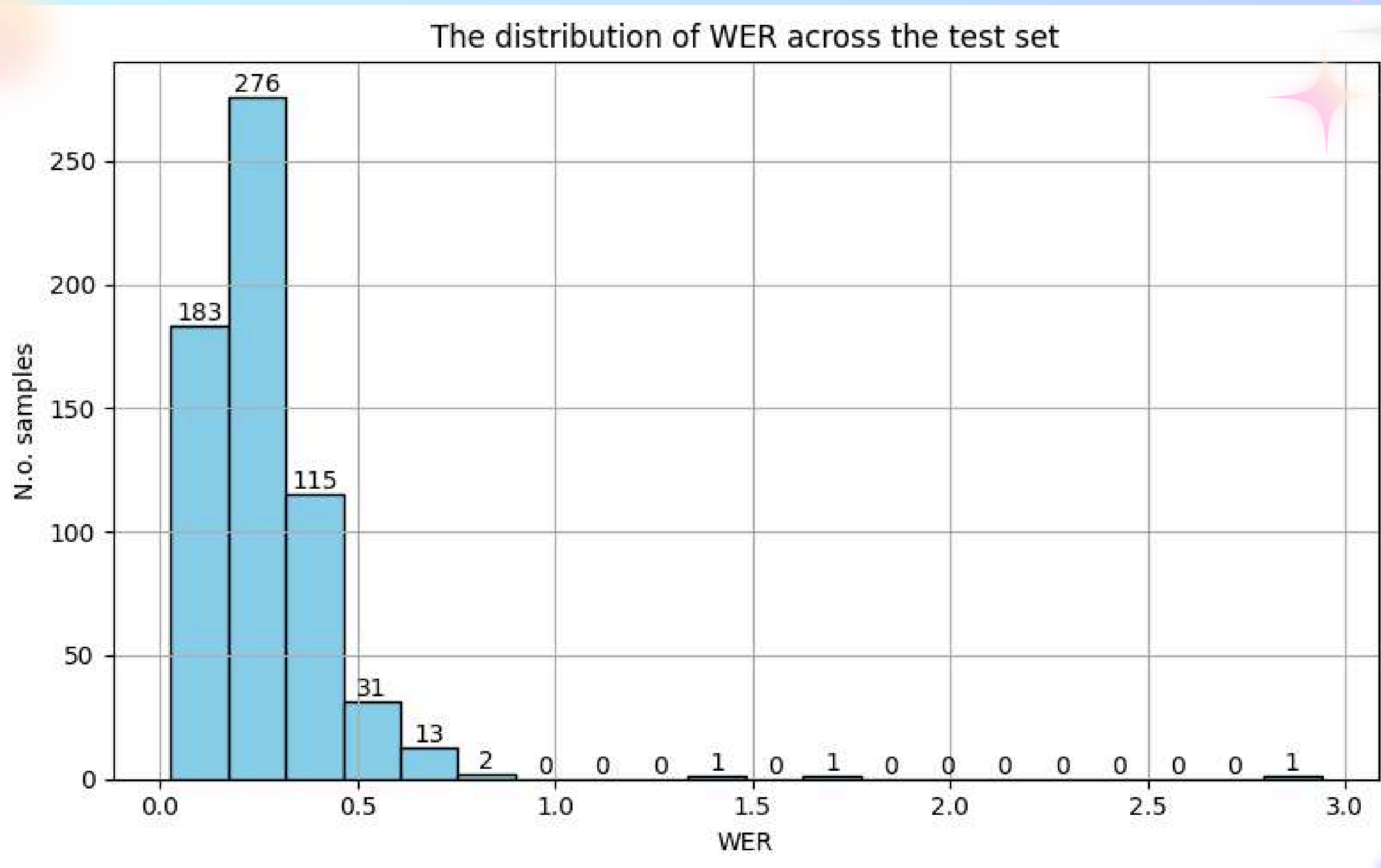


DELETE Confusions (missing)

	Count	Ground Truth	Prediction
0	164	cái	<blank>
1	101	là	<blank>
2	29	thì	<blank>
3	27	mà	<blank>
4	23	và	<blank>
5	15	nó	<blank>
6	14	ở	<blank>
7	13	rồi	<blank>
	13	cũng	<blank>
		các	<blank>

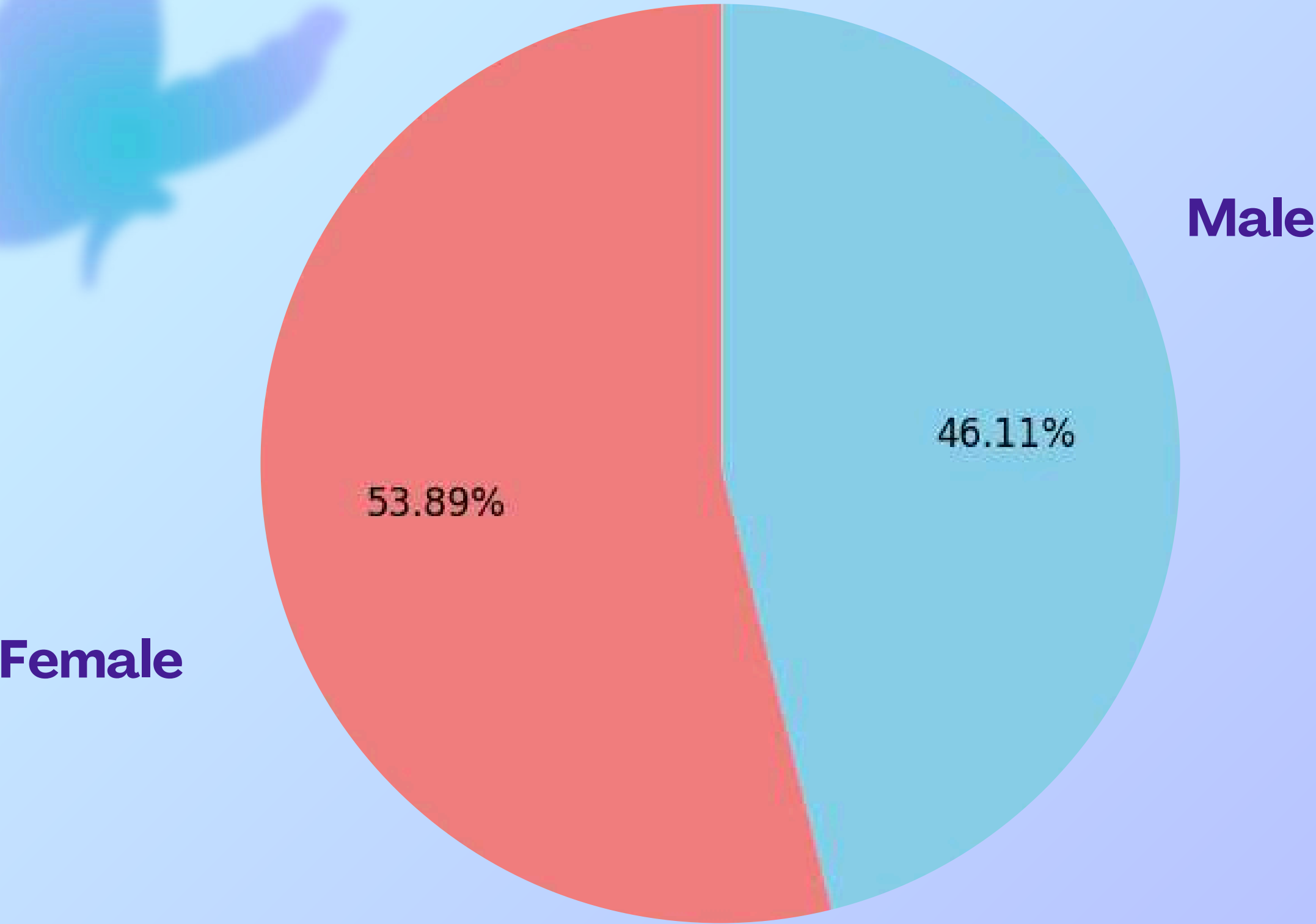


4. RESULT ANALYSIS

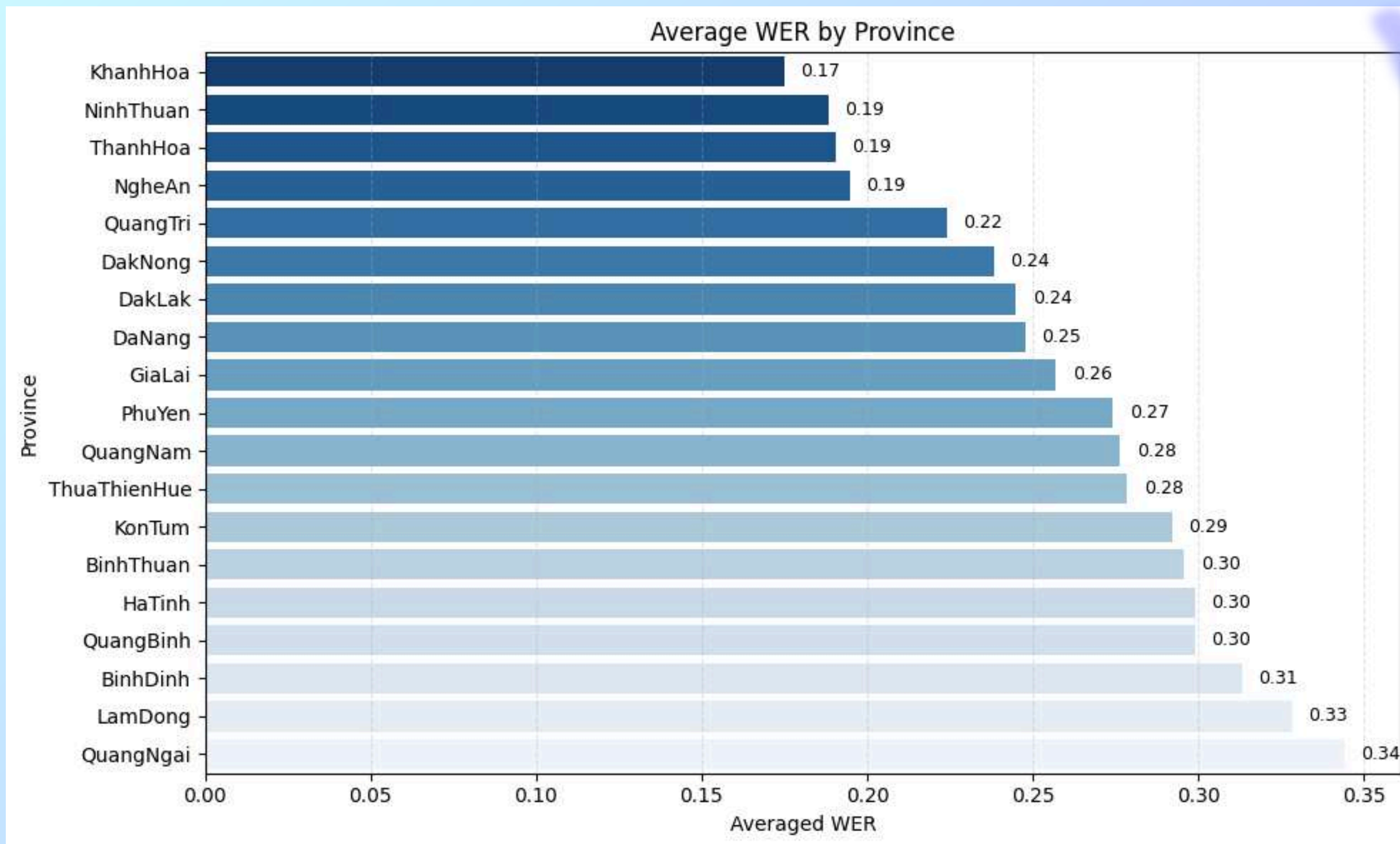


✓ WER is within acceptable range

Overall averaged WER by gender



WER by gender is relatively balanced



WER is generally lower (better) among Northern and Southern Central regions
In Nghe An, Quang Tri,... WER is better due to more data available
Lam Dong's numbers might be lower either because the data is limited, or
because it's a diverse area with people from many different regions.

Top-10 Word Confusions

+ Top INSERT Confusions (extra words)

	Count	Ground Truth	Prediction
0	305	<blank>	cái
1	180	<blank>	là
2	100	<blank>	thì
3	53	<blank>	mà
4	25	<blank>	nó
5	22	<blank>	và
6	15	<blank>	cũng
7	15	<blank>	rồi
8	14	<blank>	á
9	13	<blank>	có

✗ Top DELETE Confusions (missing words)

	Count	Ground Truth	Prediction
0	164	cái	<blank>
1	101	là	<blank>
2	29	thì	<blank>
3	27	mà	<blank>
4	23	và	<blank>
5	15	nó	<blank>
6	14	ở	<blank>
7	13	rồi	<blank>
8	13	cũng	<blank>
9	11	các	<blank>

↺ Top REPLACE Confusions

	Count	Ground Truth	Prediction
0	22	cây	cái
1	12	mà	và
2	10	đồ	rồi
3	8	tự	từ
4	7	đã	là
5	7	các	cả
6	7	con	còn
7	7	cha	cho
8	7	quang	quan
9	6	và	vào



DEMO



CONCLUSION

- Design and complete a real-world AI project with all necessary steps.
- A high-performance Central Vietnamese speech recognition system that meets expectations.

LIMITATIONS

✗ Recognition of local vocabulary is still limited (but the system performs well in recognizing speech with Central Vietnamese intonation).

✗ Lacking computational resources (GPUs)



FUTURE WORKS



Expanding the problem to include dialect normalization (local dialect → standard Vietnamese) (! due to time constraints)



Expanding the training data for provinces with limited data in the current dataset



WHAT WE LEARNT:

- Plan and execute a complete AI project
- Analyze and process data in model training
- To select, set up and train a model
- Evaluate and analyze an AI system
- Domain-specific knowledge in audio and speech-to-text technology



- Teamwork and collaboration
- Research-oriented thinking
- Self-directed learning ability
- Problem-solving skills

THANK YOU!

Ask us questions!



Special Thanks to

